Multi-chanel Smart Meter (MSM) Firmware New Features Phase II

1. Register a new device / Use an existing device
2. Automated UID Fetching
3. Data logging of Temperature
4. Resolve false values of channel
5. Implement Alerts/Alarms that has a possibility of it
    - Thresholds provided by the client:
        - Voltage unbalance (VUF) < 2 % continuous (IEEE 1159)
        - Current THD < 8 % (transformer ≤ K-factor 4), Voltage THD < 5 %
        - Sustained load < 80 % of nameplate kVA for > 30 min
        - PF > 0.95 lag (below this, schedule capacitor-bank check)
    - Status indicators on the Device provided by the client are:
        - Modem
            - Modem not responding / Modem off
            - Modem booting
            - No SIM iv. SIM Locked
            - SIM Rejected
            - Connected to Network
        - Connection Type
            - 2G / 3G / 4G
        - Signal Strength
            - None / Poor / Fair / Good / Excellent
        - MQTT Connection
            - MQTT Failed – Timed out
            - MQTT Failed - Wrong credentials
            - MQTT Failed - Connection closed or unavailable
            - MQTT Success
            - Connection lost, Reconnecting to MQTT
        - MQTTs Connection
            - MQTTs Failed - Timed out
            - MQTTs Failed - Wrong credentials
            - MQTTs Failed - Connection closed or unavailable
            - MQTTs Success
            - Connection lost, Reconnecting to MQTTs
        - OTA
            - OTA In-Progress
            - OTA Failed
            - Application already on the selected version
6. Implement login option when USB is inserted
7. When disconnection or timeout occurs, data between cloud and device should not be missed
8. Suggest & Implement OTA
9. Load data to be fetched from database for new devices
10. Remove GSM & LAN pages.

11. Add MQTTs & OTA Screens for dynamic architecture
12. Disable ports from editable
13. Implement °C & °F switch of temperature.
    When I click on the temperature it should show temperature in Fahrenheit & on click again, in Celsius like a toggle functionality.